TECH2: Mandatory assignment 3

TECH2: Introduction to Programming, Data, and Information Technology

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Mandatory assignment 3

GitHub repository: *Insert the URL to your repository here.*

About the assignment

- Complete the assignment individually.
- You may use online resources, including generative AI. Include a statement explaining how you used AI, or state that you did not use it.

Submission

- Submit your completed `assignment3.ipynb` on Canvas by **October 23, 2026, at 23:59**.

Submission requirements

- Create a new repository from the template linked on Canvas (click **Use this template** > **Create a new repository**) and implement your solution in your GitHub repository.
- Use the TECH2 environment from the course.
- Implement your solution directly in this notebook by inserting code and text cells. Do not create additional notebooks or files.
- Include all requested numerical results, figures, and short written answers.
- Give each figure a title, label its axes, and include a legend where needed.
- Before you submit, make sure your GitHub repository is **publicly accessible** and include its URL at the top of this notebook.
- Before submission, restart the kernel and run all cells in order. Your notebook must run without errors and find the CSV without an absolute file path.

Hints

- Use the material from lectures and workshops 1–3: functions, loops, NumPy, pandas, and Matplotlib.
- Compute group statistics using loops; there is no need to use `groupby()` (which we cover in lecture 4), but you can do so if you prefer.

Stock market participation in US households

Who participates in the stock market, and how has participation changed over time? To answer this question, you will analyze data from the Survey of Consumer Finances (SCF), a representative survey of US households conducted every three years from 1989 to 2022.

The file `SCF_10pct.csv` contains a 10% sample of the SCF data. The central variables of interest for this assignment are:

1. `stocks`, which reports the value of directly held stocks (in US dollars); and
2. `equity`, which reports the value of all equity holdings (in US dollars), including direct stock ownership and indirect exposure through mutual funds, retirement accounts, and other investment vehicles.

We measure stock market participation in two ways:

- **Direct participation:** a household owns individual stocks (`stocks > 0`).
- **Overall participation:** a household holds equity directly or indirectly (`equity > 0`).

For each measure, the *participation rate* is the percentage of households satisfying that condition. We therefore calculate two rates: the **direct participation rate** and the **overall participation rate**. Direct participants are included among overall participants, so the two rates should not be added.

Data description

Variable	Description
`id`	Unique observation identifier
`year`	Survey year
`age`	Age of the household reference person
`educ`	Education level: `1`: No high school; `2`: High school; `3`: Some college; `4`: Bachelor's or higher
`stocks`	Value of directly held individual stocks (in US dollars)
`equity`	Total directly and indirectly owned stock equity (in US dollars)

Part 1 — Data preparation

1. Load `SCF_10pct.csv` into a pandas DataFrame. Print the columns contained in the DataFrame and their number of non-missing values.
2. Drop all observations that have missing values for any of the variables listed in the table above. For each variable, report how many observations you dropped and how many remain.
3. Keep observations with `25 <= age <= 89`, including both endpoints. Report how many observations you dropped and how many remain.
4. Create two 0/1 indicator columns:
    - `direct`: 1 if the household participates directly (`stocks > 0`), 0 otherwise.
    - `overall`: 1 if the household participates directly or indirectly (`equity > 0`), 0 otherwise.

Part 2 — Participation rates

You are now ready to compute the stock market participation rates for the final sample. You will compute these in two ways: first, by writing a function that uses a loop to compute the participation rate, and second, by using pandas's built-in `mean()` method.

1. Write a function `participation_rate(indicator)` that takes a nonempty one-dimensional sequence of 0/1 values and returns the fraction of ones (in percent).
2. Apply your function to the final sample's `direct` and `overall` columns to compute the direct and overall participation rates. Report the results.
3. Use pandas's built-in `mean()` method to compute the direct and overall participation rates. Report the results.

Part 3 — Participation by education

1. Loop over education categories 1–4. For each category, calculate the direct participation rate and the overall participation rate (in percent).
2. Display the results, rounding rates to one decimal place.
3. Create a figure with two subplots, one for each rate, containing bar charts of participation rates across education categories.
4. In two or three sentences, describe the education pattern.

Part 4 — Participation over time

1. Loop over years. For each year, compute the direct participation rate and the overall participation rate.
2. Create a line plot showing both participation rates over time in a single figure.
3. Report the changes from 1989 to 2022 in the direct and overall participation rates, both in percentage points.
4. Describe the main time patterns in two or three sentences.

AI-use statement

Students must describe their own use of AI here, or state that they did not use it.